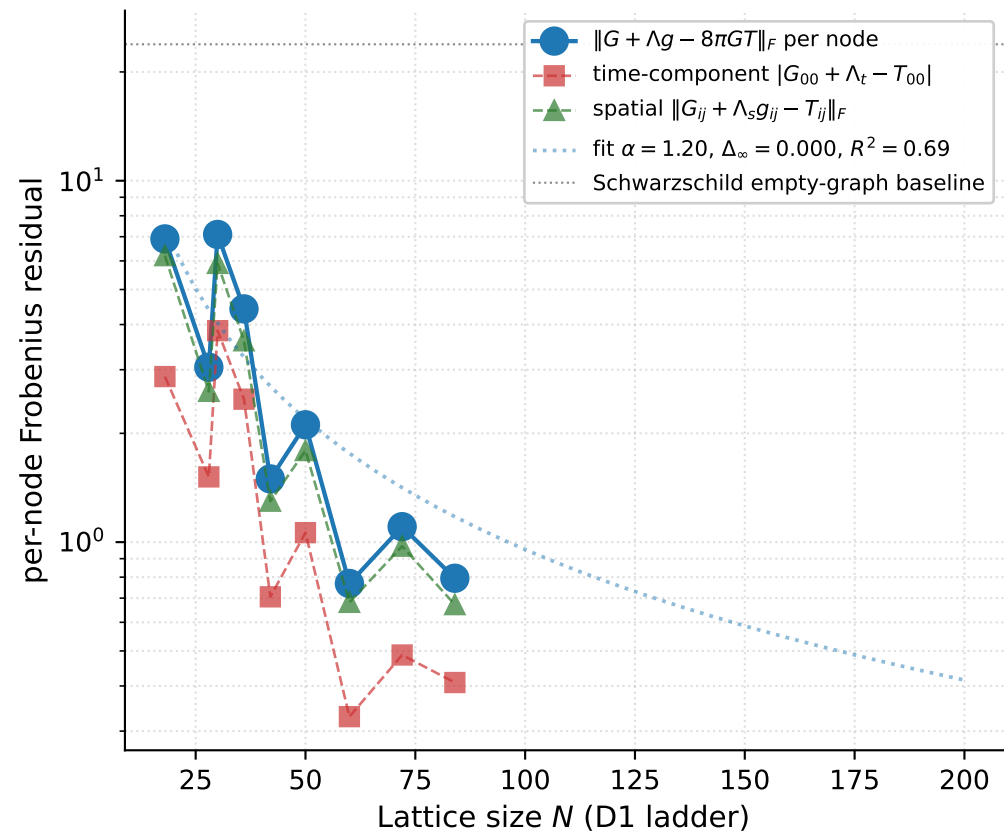


Stage B: Direct per-node 4x4 Galerkin Frobenius residual on D1 ladder (no proxies)

Direct per-node 4x4 Galerkin (Forman-Ricci, row-mean K_{rec})



Per-node Ricci scalar from Forman-Ricci spectral trace

